

Geographer + solutions engineer at the intersection of climate, geospatial, AI, and utility customers. **Overstory is the rare role where my background isn't adjacent to the job — it's inside it.** Studied GIS at Colgate; college internships at **HOT OSM, Bonobo Conservation Initiative, and oil-palm remote sensing in Indonesia** (farm boundaries, obligate forest habitats, tree counts from satellite). 3 yrs at **NPS Park Tiles** (wildfire perimeters, endangered-species habitat). 3 yrs at **Teren (formerly Solspec)** — directly relevant: **ML on drone imagery of utility right-of-ways for surface hydrology, encroachments, noxious weeds**. 2 yrs founding-advisor at Upstream AI (water-utility software). Plus Lytics (Whirlpool graph identity model). **Customer-facing by preference** — solving tech problems alongside customers is the most effective contribution I know.

01 THE FIT — JD LINE BY LINE

2–4 years with large, complex geospatial datasets in a customer- or stakeholder-facing capacity	► STRONG 3 yrs NPS (ArcGIS + Mapbox + Carto for park staff). 3 yrs Teren (geospatial-risk for utility/infra clients). 2 yrs Upstream (water-utility software for operators). Comfortably exceeds the floor.
Python-driven analytical + engineering stack — Python, Jupyter, git/GitHub, command-line workflows	► STRONG Python-first analyst daily (pandas, scikit-learn, statsmodels). Jupyter for every analysis. Git/GitHub native — sample repo: github.com/MappingKat/SunlightMtnRetail . Command-line workflows are how I move.
Vector + raster geospatial datasets — process, validate, analyze, turn into outputs	► STRONG Vector + raster is the daily reality of NPS cartography. ArcGIS, Mapbox, Carto, GeoJSON, GeoTIFF, shapefiles. Honors thesis used kriging, IDW interpolation, local Moran's I on raster + vector joins.
Translate technical findings into recommendations for non-technical customers	► STRONG NPS dashboards for rangers (non-GIS audience). Lytics Whirlpool engagement — marketing team needed to trace any segment back to its identity links + confidence scores. Upstream operator dashboards for a guy named Dale chasing a leak.
Identify patterns across customer work → convert one-off projects into repeatable artifacts	► STRONG Built the Lytics CDP delivery playbook. Built reproducible R + Snowflake AMI assessment template. Refactor notebooks → reusable scripts is the daily move.
AI tooling to amplify your work (or serious interest in)	► STRONG "Serious interest" undersells it. Founding tech advisor at Upstream AI for 2 yrs — Claude Code sub-agents, prompt libraries, RAG pipelines, embedding similarity. Would actively look for places to apply LLM-assisted analysis to the Overstory workflow.
Multiple projects · attention to detail · ambiguous problems	► STRONG Native operating mode of every startup role I've held. Comfortable with the kind of detail that survives a customer asking "where does this number come from?"
Utility operations / infrastructure / vegetation mgmt (nice to have) · Remote sensing / satellite imagery (nice to have) · SQL / data engineering (nice to have) · High-growth mission-driven startup (nice to have)	► STRONG Vegetation mgmt on utility ROWs: Teren/Solspec ML on drone imagery (encroachments, noxious weeds, surface hydrology). Remote sensing: Teren drone imagery, college oil-palm mapping in Indonesia, NPS earth-observation basemaps. Utility: 2 yrs at Upstream (water). SQL: daily (Postgres, Supabase, BigQuery, Snowflake). Mission-driven startup: only kind of company I've worked at.

02 SHIP LOG — DIRECT RELEVANCE

2012 — 15 · NATIONAL PARK SERVICE · FRONTEND / GIS · WASHINGTON, DC AREA

Team behind **NPS Park Tiles** · custom **wildfire + endangered-species mapping**

3 yrs in ArcGIS, Mapbox, Carto. On the team that built **NPS Park Tiles** — the basemap behind park websites used by millions of trip-planners. Used Park Tiles + ArcGIS to build **custom park mapping projects**: wildfire perimeters, endangered-species habitat, visitor-management decision support. **The closest analog in my background to Overstory's mission-on-public-lands flavor.**

2018 — 21 • TEREN (FORMERLY SOLSPEC) • HEAD OF MKTG TECH

Seed-stage • *ML on drone imagery of utility ROWs*

3 yrs at a geospatial-intelligence startup building **ML layers from drone imagery of utility right-of-ways** — identifying **problematic surface hydrology, encroachments, and noxious weeds**. The **closest analog to Overstory's product surface in my background**. Cross-functional w/ eng, product, sales. 4x MQLs MoM.

2024 → NOW • UPSTREAM AI • FOUNDING TECH ADVISOR

AI-native software for *rural water utilities*

Founding tech advisor. Two live Colorado pilots. **Claude Code sub-agents, RAG pipelines** for non-technical end-users. Different utility than electric, but I know utility procurement, operator language, regulatory baselines from the inside. Role is structured to step back to advisory for the right full-time seat — which this is.

2022 — 23 • LYTICS • SR. TECHNICAL CONSULTANT

Enterprise CDP • *customer-facing technical delivery*

Notable: **Whirlpool** — translated relational SQL stack into a graph identity model at scale (tens of millions of records). **Owned projects end-to-end** from data through delivery. The customer-facing technical posture this Overstory role requires.

2026 • SPRINGBOARD ML/DS CAPSTONE (R + SNOWFLAKE)

End-to-end *AMI Program Implementation Assessment*

Eagle River Valley utility — hourly interval reads on 60 meters. R + Snowflake stack. **Anomaly detection, segmentation, NRW quantification**. Same "pipeline → validate → turn into customer-actionable output" shape Overstory's workflow asks for. Published report at mappingkat.github.io/springboard-project.

EARLIER • HOMEBOT SOL ENG MGR (2015–18, THROUGH FUNDRAISE + ACQUISITION) • COLGATE HONORS THESIS (2011)

Customer delivery, data schema design, spatial statistics

Homebot: hired + managed SE team. Designed data schema for messy real-estate ingestion. Trained beta cohorts. **Honors thesis**: 180-page spatial-statistics study on 1,650 WWOOF host farms — regression, kriging, IDW, local Moran's I. Foundational systems thinking.

03 TOOLKIT

Python • pandas • scikit-learn • statsmodels

ArcGIS • Mapbox • Carto • QGIS • GeoJSON • GeoTIFF

Jupyter • git • GitHub • command-line

SQL • Postgres • Supabase • BigQuery

Claude Code • Claude API • OpenAI • Voyage • RAG

R • TypeScript • Shell

Snowflake (1 project)

CDC • event streaming • pub/sub

Plotly • D3 • Observable • Recharts

Notion • Linear • Airtable • Zapier

04 RECEIPTS — WHY THIS IS A REAL FIT

- ♦ **Geospatial depth.** 3 yrs NPS Park Tiles team — custom park mapping for **wildfire perimeters + endangered-species habitat**. Honors thesis in spatial statistics. The literal closest analog to Overstory's mission-on-public-lands flavor.
- ♦ **The Teren parallel.** 3 yrs at Teren (formerly Solspec) building **ML on drone imagery of utility right-of-ways** — surface hydrology, encroachments, noxious weeds. Same product family as Overstory. Plus college remote-sensing internship in Indonesia (oil palm + forest habitat).
- ♦ **Utility customer fluency.** 2 yrs founding-advisor at Upstream AI (small water + wastewater utilities). Different utility than electric, but I know utility procurement, operator language, regulatory baselines from the inside.
- ♦ **Customer-facing technical.** Sr. Tech Consultant at Lytics — translated tens of millions of relational records into a graph identity model for Whirlpool. Solutions Engineering Mgr at Homebot. Translating technical findings for non-technical counterparts since 2012.
- ♦ **AI by current day-job.** Founding tech advisor at Upstream AI — Claude Code sub-agents, prompt libraries, RAG, embedding similarity. The AI-tooling-to-amplify-your-work bonus is the work I'm doing now.

05 EDUCATION

BA, GIS • *Honors*

COLGATE • 2011

Honors thesis: spatial-statistics study on 1,650 WWOOF host farms — regression, kriging, Moran's I.

ML / Data Science

SPRINGBOARD • 2024

Python-first curriculum (pandas, scikit-learn, statsmodels). Capstone: R + Snowflake AMI assessment.

MBA • Full-Stack Web Dev

QUANTIC • 2021 • GALVANIZE • 2014

Strategy + finance + ops. Plus the engineering bootcamp behind every "ships code" claim.